

Ryann M. Perez

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SUMMARY

Accomplished computational biochemist with over 5 years of experience translating *in silico* experimental techniques into real-world results. Accelerated adoption of generative AI and large language models (LLMs) in biological chemistry. Harnessed machine learning (ML), simulations, and statistical analysis in collaboration with cross-disciplinary teams to unravel complex molecular mechanisms. Seeking opportunities in building novel and impactful AI systems.

EDUCATION

PhD Candidate, Computational Biochemistry

Expected March 2026

University of Pennsylvania | GPA: 3.99

Bachelor of Science, Chemistry (Minor: Biochemistry)

May 2020

University of Delaware | GPA: 3.72

RESEARCH EXPERIENCE HIGHLIGHTS

University of Pennsylvania

September 2020 - Present

Graduate Researcher in the lab of Dr. E. James Petersson

- Crafted custom LLMs to achieve state-of-the-art performance on biologic aggregation and protein stability; important findings that streamline the property prediction of proteins. Scaled LLM for inference on 64 million datapoints
- Trained students to use a novel LLM system for biological chemistry questions and answer tasks which demonstrably improved learning experiences. This technology has been adopted by faculty for recurrent use in all future classes
- Enhanced productivity by the creation of Python software packages to streamline experimental analysis, drug discovery, and protein simulations which resulted in 10 coauthored publications within 5 years

SELECTED AWARDS & PRESENTATIONS

- **Predoctoral Individual National Research Service Award (F31):** Awarded from the National Institute of Health (NIH) for an original proposal describing and prototyping an amyloid polymorphism determination assay *September 2024*
- **University of Pennsylvania Dean's Scholar:** Given to 20 students in the School of Arts and Sciences for outstanding academic achievement and intellectual promise *April 2024*
- **Lectures on Generative AI in Chemistry:** Designed and led seminars attended by over 50 professionals and scientists on utilizing ChatGPT in chemistry workflows, leading to increased awareness and implementation of generative AI within the Chemistry Department. *May 2025*

TECHNICAL SKILLS

- **Programming & ML:** Classical ML, deep learning (MLP, CNN, GCN, encoders/decoders, autoencoders, transformers, LLMs, diffusion models, contrastive learning), fine tuning, parameter efficient fine-tuning (PEFT), domain adaptation, RAG, multi-GPU and distributed training, exploratory data analysis (EDA), dimensionality reduction, unsupervised learning, dataset curation, class imbalances, experimental biases, statistical analysis. Python (PyTorch/Tensorflow, HuggingFace, scikit-learn, NumPy, Pandas). Amazon Web Services. GitHub
- **Computational Chemistry:** Custom virtual ligand screening. Simulations (Rosetta/PyRosetta). PyMOL. ADMET prediction. Structure activity relationships (SAR). Machine learning in biological contexts (Stability, solubility, aggregation). PyOpenMS
- **Chemistry:** Organic synthesis, LC-MS, HPLC, NMR, peptide synthesis, flash chromatography.
- **Biology:** Protein expression/purification, SDS-PAGE, FPLC, MALDI-TOF MS, enzymatic assays, high throughput assay design and execution, fluorescence polarization (FP).

SELECTED PUBLICATIONS

- **Perez, R. M.;** Shimogawa M.; *et al.* Large Language Models for Education: ChemTask -- An Open-Source Paradigm for Automated Q&A in the Graduate Classroom DOI: <https://arxiv.org/abs/2502.00016> *Comput. Educ.: Artif. Intel.* **Accepted**

- Li, X.; **Perez, R. M.**; Mach, R. H.; Giannakoulis, S.; Petersson, E. J. Machine Learning Prediction of Multiple Distinct High-Affinity Chemotypes for α -Synuclein Fibrils *Chem. Commun.* **2026**. DOI: 10.1039/D5CC06228D
- Li, X.; **Perez, R. M.**; *et al.* Accurate Prediction of Protein Tertiary and Quaternary Stability Using Fine-Tuned Protein Language Models and Free Energy Perturbation. *Int. J. Mol. Sci.* **2025**, 26, 7125. DOI: 10.3390/ijms26157125
- **Perez, R. M.**; Li, X.; Petersson, E. J.; Giannakoulis, S. AggBERT: Best in Class Prediction of Hexapeptide Amyloidogenesis with a Semi-Supervised ProtBERT Model. *J. Chem. Inf. Model.* **2023** DOI: 10.1021/acs.jcim.3c00817
- All publications may be found at: scholar.google.com/citations?user=nOgqddusAAAAJ&hl=en